Homework — 2.2.1 Programming Techniques

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 2.2.1 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. State the difference between **count-controlled** and **condition-controlled** iteration, and name **one** OCR construct used for each. **[2]** *(AO1)*

2. A program validates input by re-asking for a value until it is greater than 0. Explain why a `do...until` loop is more suitable here than a `while` loop. **[2]** *(AO1)*

3. Trace the call `power(2, 4)` for the recursive function below. Show the value returned at **each** level of the recursion and the final result. **[3]** *(AO2)*

```
function power(base, exp)
    if exp = 0 then
        return 1
    else
        return base * power(base, exp - 1)
    endif
endfunction
```

4. Study the program below. Note that in this language an assignment to a same-named variable inside a procedure creates a **new local** variable; it does not change the global.

```
count = 8

procedure reset()
    count = 0
    print(count)
endprocedure

reset()
print(count)
```

(a) State the two values printed, in order. **[2]** *(AO2)*

Output line 1: _ *Output line 2:* _

(b) Explain your answer to part (a), referring to **scope**. **[2]** *(AO2)*

5. A subroutine is called with `total = 10`.

```
procedure scaleByVal(byVal n)
    n = n * 2
endprocedure

procedure scaleByRef(byRef n)
    n = n * 2
endprocedure
```

State the value of `total` after each call below and justify each answer. **[2]** *(AO2)*

After `scaleByVal(total)`: total = __ **because** ____

After `scaleByRef(total)`: total = __ **because** ____

6. Define the term **encapsulation** and state **one** benefit it provides in object-oriented programming. **[1]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

7. *(2.1)* State what is meant by **abstraction** and give **one** reason it is useful when solving a problem. **[2]**

8. *(1.4.2)* A **stack** and a **queue** are both abstract data structures. State the order in which items are removed from each (use the correct three-letter acronym). **[2]**

- Stack: ______
- Queue: ______

9. *(2.1)* Explain what is meant by **decomposition**. **[2]**